Abdelrahman Khalifa

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Education

Minia University — Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering — *2020 – 2025*

- Grade: Very Good with Honors. Graduation project: AUTOSAR V2X vehicle on STM32 with a dual-bank bootloader.
- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- Treasurer, IEEE Student Branch, Minia University: managed a budget of more than $10k.

Experience

Google Summer of Code Contributor — May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) — *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI: printers grouped by device, status and supply levels, print queues, live updates, and an Add Printer flow for legacy printers (cosmic-settings #2182, in review).
- Designed it as 5 Rust crates and ran the C FFI backend as a varlink IPC service in the settings daemon, so a native crash cannot take down Settings.
- **7 PRs accepted upstream**: ported cups-rs to libcups3 with CUPS 2 and 3 builds and CI, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer — Aug 2025 – Present
EKSON — *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Developed the PyQt5 control app; used asyncio to wait for ESP32 and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Engineered remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST to relay commands and flash new firmware with esptool; GitHub Actions CI.

Embedded Linux Systems Intern — Dec 2024 – Jan 2025
Orange Digital Center (ODC) — *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* — Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

IoT Device Communication System | *C++, Raspberry Pi 4, QEMU, Yocto* — GitHub

- Object-oriented framework for TCP unicast and UDP multicast between a server and emulated clients.

Operating System Simulator | *Java* — GitHub

- CLI simulating scheduling, memory management, system calls and locks.

Inventory Management System | *C#, WinForms, SQL* — GitHub

- Stock, orders and deliveries app with role-based access control and reports.

Technical Skills

Languages: Rust, Modern C++, C, Python (asyncio), JavaScript (Node.js), SQL, Java, C#, Bash
Computer science: Data structures and algorithms, operating systems, concurrency and multithreading, async/event-driven design, memory management, networking (TCP/UDP, IPC)
Backend & tools: Node.js, Express, REST APIs, MongoDB (NoSQL), PyQt5; Git (version control), GitHub Actions CI, CMake, testing, debugging and troubleshooting, code review